Toward Generalizable Whole-Body Skill Learning

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***Abstract*— Humanoid robot learning has recently expanded beyond motion imitation and locomotion toward more general-purpose whole-body competence, including modular skill composition, vision-language grounded control, and dynamic loco-manipulation. To deepen the understanding of high-dimensional control in such systems, this report reviews three recent studies from complementary perspectives: (1) learning compositional part-level skills for scalable motion control, (2) integrating vision and language with hierarchical latent whole-body control, and (3) combining reinforcement-learned whole-body policies with online planning for dynamic manipulation. Through this overview, we highlight key insights for designing effective learning and control strategies in complex, multi-DoF robotic systems.**

I. Introduction

Recent advances in large-scale simulation, motion datasets, and multimodal learning have accelerated progress in humanoid control. In particular, recent studies have moved from narrow task-specific controllers toward frameworks that aim to support reusable skills, semantic instruction following, and robust adaptation to diverse environments. These trends are especially relevant to high-degree-of-freedom robotic systems, where coordinated whole-body behavior must emerge from efficient representations and structured control architectures.

Our research interest is rooted in modular robotics. Modular robots offer flexibility, reconfigurability, and scalability, but they also introduce significant control challenges because coordinated behavior must be achieved across many interacting components. From this viewpoint, humanoid robots provide a useful reference platform: they require rich whole-body coordination, integrate perception and control, and increasingly rely on hierarchical learning structures that resemble the organization needed in modular systems.

To deepen our understanding of these design principles, this reading seminar focuses on three recent papers on humanoid or whole-body control from complementary perspectives. Specifically, we examine: (1) modular skill learning for scalable motor representation, (2) latent vision-language instruction for humanoid whole-body control, and (3) hierarchical reinforcement learning and model-predictive planning for dynamic loco-manipulation. Through this overview, we aim to extract principles for handling high-dimensional, coordinated control and to relate these findings back to the study of modular robotic systems.

II. ModSkill: Physical Character Skill Modularization [1]

A. Background

Large-scale human motion datasets have enabled imitation learning methods to acquire diverse motor skills for simulated humanoids and physical characters. However, existing approaches often rely either on a universal full-body tracking controller or on a single latent skill space for the entire body. Although these approaches can reproduce many motions, they often struggle to scale to larger datasets and to generalize across highly diverse behaviors. In particular, a single full-body representation may become too coarse or too difficult to optimize when the motion space grows in complexity.

This motivates the need for a more structured representation of motor behavior. The central idea of ModSkill is that human motion is inherently modular: different body parts can express semi-independent yet composable functions. If full-body motion can be decomposed into part-specific skills, then learning may become more scalable, and the resulting representations may become more reusable for downstream control tasks.

B. Method

The paper proposes a modularized skill learning framework that decouples full-body motion into compositional skills for individual body parts. Instead of representing motion with a single latent variable for the entire body, the policy uses a *skill modularization attention layer* that transforms observations into separate modular skill embeddings, each associated with a specific body region such as the arms, legs, or torso.

These modular embeddings are then provided to independent low-level controllers for each body part. In this way, the controller architecture explicitly reflects the decomposition of full-body behavior into coordinated part-level policies. The learned modular skill spaces are compact and compositional, enabling different part-wise skills to be combined into diverse full-body behaviors.

In addition, the paper introduces an *Active Skill Learning* scheme with *Generative Adaptive Sampling*. Rather than relying only on a fixed motion dataset, the method uses a pre-trained motion generation model to synthesize additional training examples for difficult motion segments. This improves coverage of challenging tracking scenarios and helps the policy learn more robust part-wise skills.

After training, the modular low-level skills can be reused by a high-level task policy for downstream tasks, allowing

the framework to transfer learned motion knowledge without requiring a separate full-body distillation stage.

C. Experiments and Results

The method is evaluated on full-body motion tracking as well as downstream goal-driven tasks such as steering, reaching, striking, and VR tracking. Experimental results show that the modularized skill learning framework improves precise motion tracking and achieves strong downstream performance. Compared with prior methods such as ASE and ASE-PMP, ModSkill attains higher normalized returns on steering and reaching tasks, and performs competitively with strong baselines such as PULSE while avoiding the need for an additional distillation pipeline.

The learned modular embeddings also exhibit interpretable structure. Skill interpolation experiments show that smoothly changing the embedding for a single body part results in smooth local behavioral changes while preserving the rest of the body motion. Likewise, skill composition experiments demonstrate that independent part-wise skills can be combined to synthesize coordinated full-body poses and behaviors. Overall, the results suggest that modular skill decomposition can improve scalability, reusability, and controllability in complex humanoid motion learning.

III. LeVERB: Humanoid Whole-Body Control with Latent Vision-Language Instruction [2]

A. Background

Vision-language-action models have recently shown strong semantic reasoning and zero-shot generalization in manipulation and navigation. However, most prior systems assume a simple low-level action interface, such as end-effector poses or base velocities. While such interfaces are convenient for quasi-static tasks, they are often too limited for agile humanoid whole-body control, where actions must capture coordinated full-body motion and contact-rich scene interaction.

This creates a gap between high-level multimodal understanding and low-level whole-body execution. A humanoid robot needs not only semantic grounding from vision and language but also an action interface expressive enough to support diverse whole-body behaviors. The paper addresses this challenge by introducing a latent action representation that bridges vision-language reasoning and dynamic humanoid control.

B. Method

The paper proposes LeVERB, a hierarchical latent instruction-following framework for humanoid vision-language whole-body control. The framework has two major components. At the high level, *LeVERB-VL* processes visual observations and language instructions to predict a structured latent action distribution, referred to as a latent action vocabulary or latent verbs. At the low level, *LeVERB-A* consumes sampled latent codes together with proprioceptive observations and outputs dynamics-level control actions for the humanoid.

To train the system, the authors first build *LeVERB-Bench*, a sim-to-real-ready benchmark with more than 150 tasks from 10 categories. Since large-scale visual humanoid demonstrations are difficult to collect on real hardware, they construct a synthetic data generation pipeline by replaying retargeted humanoid motions in photorealistic IsaacSim scenes and annotating them with language instructions. This produces paired image, text, and motion trajectories for training.

LeVERB-VL is trained using a kinematic reconstruction objective with a CVAE-style latent structure so that the latent space aligns visual-language semantics with whole-body motion patterns. Then, the latent space is frozen, and LeVERB-A is trained by distilling teacher policies into a unified student controller conditioned on latent samples. This decoupled training strategy allows the high-level module to focus on semantics and the low-level module to focus on stable whole-body dynamics.

C. Experiments and Results

The proposed framework is evaluated on a range of multimodal humanoid tasks in LeVERB-Bench, including visual navigation, locomotion, reaching, and sit-down behaviors. Experimental results show that LeVERB achieves strong zero-shot performance, reaching 80% success on simple visual navigation tasks and about 58.5–58.9% average success overall, substantially outperforming a naive hierarchical baseline.

Ablation studies indicate that the structured latent representation and latent sampling process are both important. Removing the high-level vision-language planning capacity or using an unstructured deterministic latent space leads to major performance drops, especially on visual-language tasks that require semantic grounding and spatial reasoning. The model also shows robustness to vocabulary variation and can adapt behavior based on scene geometry, such as sitting differently depending on the location of a chair.

Finally, the framework is deployed zero-shot on a real humanoid robot, demonstrating that synthetic-data-trained latent vision-language control can transfer to hardware without task-specific retraining. Overall, the results suggest that a latent multimodal interface is a promising way to integrate semantic reasoning with agile whole-body humanoid control.

IV. SUMO: Dynamic and Generalizable Whole-Body Loco-Manipulation [3]

A. Background

Loco-manipulation requires robots to coordinate locomotion and object interaction simultaneously. This becomes especially challenging when the manipulated objects are large, heavy, or geometrically complex, since success then requires dynamic whole-body coordination rather than isolated arm motion. Traditional reinforcement learning approaches can learn robust behaviors, but they often require extensive reward engineering and retraining for each new manipulation task. Conversely, sample-based model-predictive control

is flexible and training-free, but it struggles with high-dimensional action spaces and unstable legged dynamics.

This motivates a hybrid framework that can combine the robustness of learned whole-body control with the adaptability of online planning. The key question is how to preserve the flexibility of model-based search while reducing the difficulty of planning in a full humanoid or quadruped action space.

B. Method

The paper proposes *Sumo*, a hierarchical framework that combines a pre-trained generalist whole-body control policy with a high-level sample-based model-predictive controller. The low-level policy is learned with reinforcement learning and takes the current state together with desired torso, arm, and leg commands, outputting joint-level motor commands at high frequency. The high-level controller runs online and searches over a compact command space to steer this policy toward task completion.

A central design choice is the use of *policy-in-the-loop rollouts*. Instead of planning directly in the robot's raw joint action space, the model-predictive controller performs rollouts through the low-level whole-body policy. This reduces the effective dimensionality of the search problem and leverages the low-level controller as a stabilizing prior. As a result, online planning can focus on task-relevant whole-body command variables and on manipulation objectives rather than locomotion stabilization itself.

Because the planner operates at test time, the same learned controller can be reused for new objects and task goals by changing the object model or cost function, without retraining. This provides a mechanism for generalization that complements the robustness of the pre-trained whole-body policy.

C. Experiments and Results

The framework is evaluated on challenging whole-body loco-manipulation tasks involving objects near or beyond the robot's nominal manipulation capacity. On a real Spot quadruped robot, Sumo successfully performs tasks such as uprighting tires and chairs, stacking tires, dragging crowd-control barriers and tire racks, and pushing a large box. Reported success rates are high across these tasks, typically around 8 to 10 successful trials out of 10 depending on the object and scenario.

The same framework is also applied to humanoid loco-manipulation in simulation using a Unitree G1 robot, where it solves tasks such as box pushing, door opening, chair pushing, and table pushing. Experimental comparisons indicate two main advantages of the hierarchical design. First, it simplifies loco-manipulation relative to end-to-end MPC and end-to-end RL by reducing search difficulty and alleviating reward-design complexity. Second, it improves generalization because new tasks can be handled by updating the planner's objective at deployment time rather than retraining the whole controller.

Overall, the results demonstrate that combining a robust low-level whole-body policy with online planning is an effective strategy for dynamic and generalizable loco-manipulation in high-dimensional robots.

V. Conclusion

This report reviewed three recent studies on whole-body control from the perspectives of modular skill learning, multimodal instruction following, and hierarchical loco-manipulation. Although the technical settings differ, the papers share a common theme: complex robot behavior becomes more tractable when control is organized through structured intermediate representations rather than learned end to end in a flat manner.

First, ModSkill showed that part-wise decomposition can provide compact and reusable motor representations, improving both scalability and compositionality. Second, LeVERB demonstrated that a structured latent interface can bridge vision-language understanding and dynamics-level humanoid control, enabling semantically grounded whole-body behaviors. Third, Sumo illustrated that combining a learned whole-body controller with online planning can achieve flexible and generalizable dynamic manipulation without retraining for every task.

From the perspective of modular robotics, these works suggest several useful principles. Structured decomposition can reduce the difficulty of learning and improve reuse. Hierarchical interfaces can connect high-level reasoning with low-level execution. Finally, combining learned policies with online adaptation can improve robustness in diverse and changing environments. These principles are likely to be valuable not only for humanoids but also for modular robotic systems, where many components must cooperate to produce coherent whole-body behavior.

References

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